QUBO Formulation: DNA Read Ordering Problem for Genome Assembly

DNA Read Ordering Problem for Genome Assembly

Problem Statement. Given a set of N DNA reads and an $N \times N$ pairwise overlap matrix O, where $O_{i,j}$ denotes the length of the longest suffix of read i that matches a prefix of read j, the objective is to determine a linear ordering of all reads that maximizes the total overlap between consecutive reads in the sequence. Formally, we seek a permutation π of $\{0, \ldots, N-1\}$ that maximizes $\sum_{k=0}^{N-2} O_{\pi(k),\pi(k+1)}$.

Decision Variables. For each read index $i \in \{0, \ldots, N-1\}$ and each position index $k \in \{0, \ldots, N-1\}$, we introduce a binary decision variable $x_{i,k} \in \{0, 1\}$. The variable $x_{i,k}$ equals 1 if and only if read i is assigned to position k in the final ordering; otherwise, $x_{i,k} = 0$. The total number of variables is N^2.

QUBO Derivation. **Step 1.** The original objective is to maximize the sum of consecutive overlaps. To cast this as a minimization problem suitable for QUBO, we negate the objective function:

$$H_{\text{obj}}(x) = -\sum_{k=0}^{N-2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} O_{i,j}\, x_{i,k}\, x_{j,k+1}. \tag{1}$$

Step 2. The assignment must satisfy two sets of equality constraints. First, each read must appear exactly once:

$$\sum_{k=0}^{N-1} x_{i,k} = 1, \quad \forall i \in \{0, \ldots, N-1\}. \tag{2}$$

Second, each position must be occupied by exactly one read:

$$\sum_{i=0}^{N-1} x_{i,k} = 1, \quad \forall k \in \{0, \ldots, N-1\}. \tag{3}$$

Step 3. We convert the equality constraints into quadratic penalty terms using a positive weight λ. For the read-assignment constraints, the penalty for a fixed i is:

$$\lambda \left(\sum_{k=0}^{N-1} x_{i,k} - 1 \right)^2 . \tag{4}$$

Expanding the square and applying the idempotent property $x_{i,k}^2 = x_{i,k}$ for binary variables yields:

$$\lambda \left(\sum_{k=0}^{N-1} x_{i,k}^2 + \sum_{k \neq l} x_{i,k} x_{i,l} - 2 \sum_{k=0}^{N-1} x_{i,k} + 1 \right) = \lambda \left(1 - \sum_{k=0}^{N-1} x_{i,k} + \sum_{k \neq l} x_{i,k} x_{i,l} \right) . \tag{5}$$

An identical expansion applies to the position-assignment constraints for a fixed k:

$$\lambda \left(\sum_{i=0}^{N-1} x_{i,k} - 1 \right)^2 = \lambda \left(1 - \sum_{i=0}^{N-1} x_{i,k} + \sum_{i \neq j} x_{i,k} x_{j,k} \right) . \tag{6}$$

Step 4. Combining the objective and all penalty terms gives the full QUBO Hamiltonian $H(x)$:

$$\begin{aligned} H(x) = &- \sum_{k=0}^{N-2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} O_{i,j}\, x_{i,k}\, x_{j,k+1} \\ &+ \lambda \sum_{i=0}^{N-1} \left(1 - \sum_{k=0}^{N-1} x_{i,k} + \sum_{k \neq l} x_{i,k} x_{i,l} \right) \\ &+ \lambda \sum_{k=0}^{N-1} \left(1 - \sum_{i=0}^{N-1} x_{i,k} + \sum_{i \neq j} x_{i,k} x_{j,k} \right) . \end{aligned} \tag{7}$$

Step 5. Collecting linear and quadratic terms, and using the symmetric QUBO representation $H(x) = \sum_{p \leq q} Q_{p,q} x_p x_q + c$, we read off the matrix entries and offset. Variables are indexed by pairs (i, k). The Q matrix entries are:

$$Q_{(i,k),(j,l)} = \begin{cases} -O_{i,j} & \text{if } l = k + 1 \text{ and } 0 \leq k \leq N - 2, \\ -\lambda & \text{if } i = j \text{ and } k = l, \\ 2\lambda & \text{if } i = j \text{ and } k \neq l, \\ 2\lambda & \text{if } k = l \text{ and } i \neq j, \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

$$c = 2N\lambda. \tag{9}$$

Penalty Weight Justification. The penalty weight λ must be chosen strictly greater than the maximum magnitude of any overlap score: $\lambda > \max_{i,j} |O_{i,j}|$. This condition guarantees that any violation of the assignment constraints incurs a penalty energy strictly larger than the maximum possible reduction in the objective function. Since the objective coefficients are bounded by $\max |O_{i,j}|$, a larger λ ensures that the energy landscape penalizes infeasible configurations more heavily than any feasible configuration can benefit from the objective terms.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{N^2}$ recovers the optimal read ordering because feasible solutions satisfy all constraints, causing all penalty terms to vanish and yielding $H(x) = H_{\text{obj}}(x)$. Infeasible solutions violate at least one constraint, incurring a penalty of at least λ. By the choice $\lambda > \max |O_{i,j}|$, the energy of any infeasible configuration strictly exceeds the energy of the best feasible configuration. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that minimizes $H_{\text{obj}}(x)$, which is equivalent to maximizing the original overlap sum.

Illustrative Example. Consider a concrete instance with $N = 3$ reads. The overlap matrix is:

$$O = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 3 \\ 2 & 1 & 0 \end{pmatrix}.$$

The 9 binary variables are linearly mapped to indices $0, \ldots, 8$ via $x_{i,k} \mapsto x_{3i+k}$. Substituting $N = 3$ and the matrix entries into the general formulas yields the concrete Hamiltonian:

$$\begin{aligned} H(x) = &-2x_{0,0}x_{0,1} - x_{0,0}x_{0,2} - x_{0,1}x_{1,2} - 3x_{1,1}x_{1,2} - 2x_{2,0}x_{2,1} - x_{2,0}x_{2,2} \\ &+ \lambda \sum_{i=0}^{2} \left(1 - \sum_{k=0}^{2} x_{i,k} + \sum_{k \neq l} x_{i,k}x_{i,l}\right) + \lambda \sum_{k=0}^{2} \left(1 - \sum_{i=0}^{2} x_{i,k} + \sum_{i \neq j} x_{i,k}x_{j,k}\right). \end{aligned} \tag{10}$$

Using the Q matrix structure, the non-zero off-diagonal objective terms correspond to $Q_{0,1} = -2$, $Q_{0,2} = -1$, $Q_{4,5} = -3$, $Q_{6,7} = -2$, $Q_{6,8} = -1$ (all others zero). The diagonal entries are $Q_{p,p} = -\lambda$ for $p \in \{0, \ldots, 8\}$, and all same-row/same-column off-diagonals are 2λ. The constant offset is $c = 6\lambda$.

For the feasible assignment corresponding to the ordering $r_0 \to r_1 \to r_2$, we set $x_{0,0} = 1$, $x_{1,1} = 1$, $x_{2,2} = 1$ and all other x to 0. The penalty terms evaluate to zero. The objective energy is $H_{\text{obj}} = -(O_{0,1} + O_{1,2}) = -(2 + 3) = -5$. Thus, $H(x) = -5 + 6\lambda$. Any permutation violating the assignment constraints would incur a penalty of at least λ, resulting in an energy $\geq -5 + 7\lambda > -5 + 6\lambda$ (since $\lambda > 3$). Hence, the global minimum correctly identifies the optimal ordering.